Amritha Pradeep

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Profile

Currently pursuing M.Sc. Mathematics at IIT Madras with strong academic foundations in pure mathematics and a growing interest in coding theory and cryptography. Developed early exposure to mathematical research through guided reading projects and advanced mathematical camps. Actively seeking deeper engagement with research in coding theory and cryptography.

Education

Indian Institute of Technology Madras, Chennai
M.Sc. Mathematics (2025–2027)
CGPA (First and second semester combined): 8.92

Stella Maris College, Chennai
B.Sc. Mathematics (2022–2025)
CGPA: 9.8
Secured First Rank

Delhi Private School, Sharjah
Class XII: 97.6%
Class X: 96%

Research Interests

Number Theory • Coding Theory • Cryptography

Reading Projects & Academic Exposure

Reading Project under Dr. Ananthnarayan Hariharan — IIT Bombay (December 2025)
Studied foundational algebraic structures including fields and field extensions, before moving to introductory coding theory. Explored basic error-correcting codes, Hamming distance, and related concepts in coding theory.

IMSc Summer Research Program — Institute of Mathematical Sciences, Chennai (May–June 2026)
summer research program under Professor Sanoli Gun where I explored introductory topics in algebraic number theory, including Legendre symbols and p-adic integers, through a guided reading of A Course in Arithmetic by J.P. Serre

Academic Programs & Mathematical Camps

- MTTS Level-O Summer Camp (2024)
- MTTS Level-I Summer Camp (2025)
- Mentor at OFCM by MTTS Trust (2024, 2025)
- Madhava Mathematics Nurture Camp:
  - Chennai Mathematical Institute (2024)
  - IIT Bombay (2025)
- Vigyan Vidhushi Program 2025 at TIFR Mumbai

Courses & Certifications

NPTEL

- Advanced Engineering Mathematics — IIT Kharagpur (Course Topper)
- Basic Calculus — IIT Madras
- The Joy of Computing using Python — IIT Ropar

Coursera

- Introduction to Mathematical Thinking — Stanford University
- Linear Algebra: Orthogonality and Diagonalization — Johns Hopkins University

Technical Skills

Python • LaTeX

Languages

English • Tamil • Hindi • Malayalam • Arabic